

Chapter 3

CS1 course package - Knox College (KC)

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Abstract

|Your abstract text here.|

Descriptor Elements

Textbook used: None

Programming Language Employed: Java

Relevant core courses: CS1

Pre-requisites: None

Relevant PDC topics: Classification of PDC Topics according to Bloom's Taxonomy[1]: Remember (RE), Understand (UN), Apply (AP), Analyze (AN), Evaluate (EV), and Create (CR)

PDC Concept	Chapter Section				
	1.1	1.2	1.3	1.4	1.5
Topic1	K	C	A		
Topic2	K	C	A		
Topic3		C	A	A	

Learning outcomes:

1. |Outcome 1|
2. |Outcome 2|

Context for use: By way of context, Knox College is a small, residential liberal arts institution. It uses an academic calendar with 3 terms (Fall, Winter, and Spring), each slightly less than 10 weeks long. The CS2 class meets 4 times a week, 3 lectures (MWF) and a lab on either Tuesday or Thursday (which depends on the term, but all labs are on the same day each term). All class meetings are 70 minutes long. The intent is that a Knox course should be equivalent to a semester course at other institutions.

The size of this class varies, but the targeted size is 24 students. The class is the first course in the Computer Science major, the Computer Science minor, and the Data Science major. It is also required for students majoring in Biology and Neuroscience who complete a BS instead of a BA in those fields. The course is also a common option for students in the Business & Management major and is taken by many students as an elective. Historically, roughly half of the students from this course continue to CS2.

This course uses Peer Instruction [3, 4], a pedagogy where lecture is periodically interrupted with a multiple-choice question. The students answer these individually, then discuss them in groups and re-answer. Then the instructor reveals the correct answer, has a student explain this answer, and then fills in any additional material. The in-class questions are graded only for participation. Lab assignments are homework; the

students begin the assignments in lab, when the instructor and a TA are available to answer questions. This gets them over the hurdle of starting the assignment. Then they have until the beginning of the next lab (a week) to complete the assignment.

3.1 Introduction

|Motivation and background for the chapter.|

3.2 How CS1 was changed and PDC topics integrated

|Explain how the existing course (e.g., CS1) was modified.|

3.2.1 Syllabus

Here is the new course calendar at a week-by-week level:

Week	Topics
1	Course introduction; types and variables
2	Conditionals; strings; loops
3	Methods; data parallelism
4	Return values; functions (static methods)
5	Arrays; ArrayList; identifying mystery functions
6	Classes and objects
7	Event-driven programming
8	Connecting to remote server
9	Recursion

Day-by-day level detail as well as the specific lecture materials and assignments are provided in this chapter's appendix.

3.2.2 Highlights

|What remained the same, and what specifically needed to be changed or deleted?|

3.2.3 Challenges

|Discuss logistical, pedagogical, or technical difficulties.|

3.3 New Unplugged Activities

3.3.1 Identifying mystery functions

Problem Description, Prerequisites, and Learning Outcomes

Implementation Details

- **Logistics** (Material required for the activity, etc.)
- **How to Conduct**
- **Instructional material** (Lectures, Videos, Assignments, etc.)

Experience and Teaching Tips

Data and Analysis

3.4 New Plugged-in Activities

3.4.1 Programming Flagmaker

Problem Description, Prerequisites, and Learning Outcomes

Algorithms

Implementation Details

- Instructional Material (Lectures, Videos, Assignments, Tests)
- How-To Guides

Experience and Teaching Tips

Data and Analysis

3.4.2 Knoxcraft

Problem Description, Prerequisites, and Learning Outcomes

[2]

Algorithms

Implementation Details

- Instructional Material (Lectures, Videos, Assignments, Tests)
- How-To Guides

Experience and Teaching Tips

Data and Analysis

3.5 Course-wide Pre and Post Course Surveys

We gave pre- and post-surveys for this course during each term that it was offered, but haven't analyzed the data yet.

3.6 Other Activities and Ideas

[Alternative ways to apply these concepts, etc..]

3.7 Conclusion

Bibliography

- [1] Lorin W. Anderson and David R. Krathwohl, editors. A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom’s Taxonomy of Educational Objectives. Longman, New York, NY, 2001.
- [2] M. Gerten, J.M. Iqrah, and J. Spacco. Knoxcraft Terps: Parallel Minecraft for novices, a Peachy Assignment. In Proc. 16th NSF/TCPP Workshop on Parallel and Distributed Computing Education (EduPar), 2026.
- [3] L. Porter, C.B. Lee, and B. Simon. Halving fail rates using peer instruction: a study of four computer science courses. In Proc. 44th SIGCSE Technical Symp. Computer Science Education (SIGCSE TS), pages 177–182, 2013.
- [4] Leo Porter and Beth Simon. Retaining nearly one-third more majors with a trio of instructional best practices in CS1. In Proc. 44th SIGCSE Technical Symp. Computer Science Education (SIGCSE TS), pages 165–170, 2013.

Appendix

|You must include these four specific subsections (in addition to anything else):|

Github Link to Instructional Material

`https://github.com/...`

Detailed Pre and Post Syllabus

Organization of Material

|Explain the folder structure of your Github repository.|

Survey and Other Anonymized Data and Analysis

|Include student feedback, learning data, or implementation analysis.|